EAS439

Examination - January Semester 2024

Numerical Analysis

Tuesday, 16 April 2024 **01:00 pm - 03:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FOUR (4)** questions and comprises **SEVEN (7)** printed pages (including cover page).
2. You must answer **ALL** questions.
3. All answers must be written in the answer book.
4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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Answer all questions. (Full marks: 100)

Question 1a

Figure Q1a shows the forces acting on an aircraft at level flight.

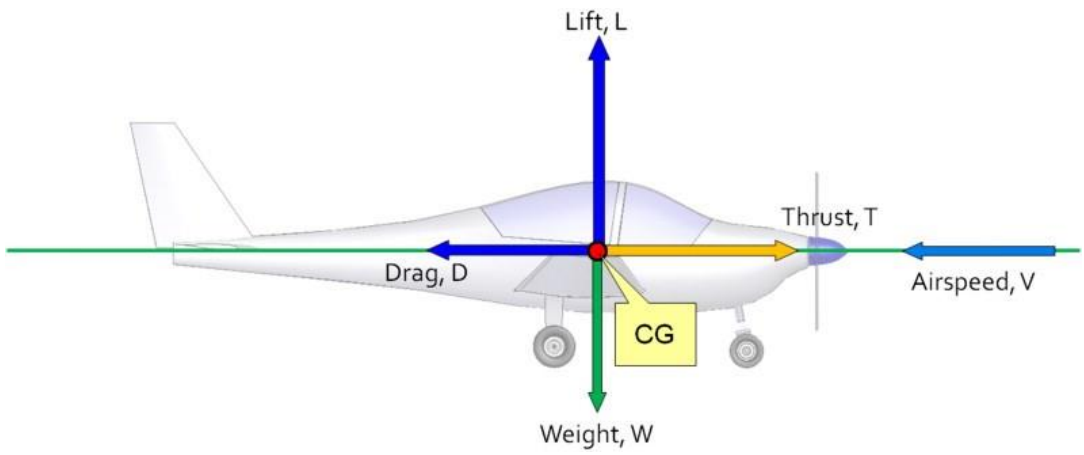

Figure Q1a

From Newton's 2nd law, the governing equation is given by:

$$\frac{W}{g}\frac{dV}{dt} = T - D = T - \left[C_{D0} + k\left(\frac{W}{\frac{1}{2}\rho V^2 S}\right)^2\right](\tfrac{1}{2}\rho V^2 S)$$

where C_{D0} and k are the parameters for the aircraft aerodynamic drag polar, S is the wing area, and ρ is the air density.

Question 1a(i)

Formulate numerical approximations to the equation using forward difference to compute the aircraft speed V, for the level flight to accelerate speed from V_1 to V_2.

(8 marks)

Question 1a(ii)

An aircraft has the following characteristics:

Mass: $m = 10{,}000\ kg$

Drag Polar: $C_{D0} = 0.02$, $k = 0.04$

Wing Area: $S = 60\ m^2$

Thrust: $T = 9{,}000\ N$

Air density: $\rho = 1.225\ kg/m^3$

Initial Speed: $V_1 = 65\ m/s$

Based on the numerical approximation formulated in Question 1a(i), compute the aircraft speed V at different time, using Δt = 5. Copy Table Q1a(ii) to answer the question.

Time (s)	aircraft speed V (m/s)
0	65
5	
10	
15	

Table Q1a(ii)

(9 marks)

Question 1a(iii)

Analyse the accuracy and efficiency of the numerical formulation used, by commenting the effect of Δt on the result accuracy and the algorithm efficiency.

(3 marks)

Question 1b

Explain the difference between "Total Error" and "Approximation Error", and their respective usage.

(5 marks)

Question 2a

Given the following system of equations:

$$\begin{aligned} 2x_1 + x_2 - x_3 &= 2 \\ 5x_1 + 2x_2 + 2x_3 &= 9 \\ 3x_1 + x_2 + x_3 &= 5. \end{aligned}$$

Calculate the solution using Gauss-Jordan Elimination method. Show all steps of the computation.

(15 marks)

Question 2b

The deflection of a cantilever beam under the uniformly distributed load (w) is given as follow:

$$\frac{dy}{dx} = \frac{w}{24EI}(4x^3 - 12Lx^2 + 12L^2x),$$

where y is the vertical deflection, E is the modulus of elasticity, I is the area moment of inertia, and L is the length of the beam. The cantilever beam is fixed at root $x = 0$ and the free end is at $x = L$. Use Simpson's 3/8 rule to design and implement a computational step to calculate the deflection of the beam at $x = L$.

(10 marks)

Question 3

A property ϕ is transported by means of convection and diffusion through the one-dimensional domain as shown in Figure Q3.

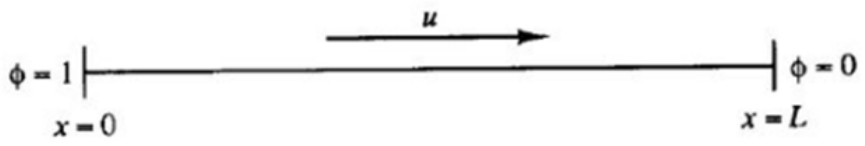

Figure Q3

The steady state governing equation is given as follow:

$$\frac{d}{dx}(\rho u \phi) = \frac{d}{dx}\left(\mu \frac{d\phi}{dx}\right)$$

where u is the stream-wise flow velocity and μ is the coefficient of viscosity.

Question 3a

Formulate a control volume discretization scheme, for the flow problem described in Figure Q3, by dividing the entire domain into N_x number of cells. Indicate clearly the followings:

i. Size of each finite volume cells;
ii. Location where the property ϕ is defined;
iii. Propose how boundary conditions are to be defined;
iv. Define the control surfaces; and
v. Numerical approximation of the fluxes at the control surfaces.

(10 marks)

Question 3b

Formulate the numerical expression for the boundary condition (at $x = 0$ and $x = L$) to be used.

(6 marks)

Question 3c(i)

Appraise the accuracy of the numerical scheme.

(4 marks)

Question 3c(ii)

Propose **THREE (3)** methods to verify and validate the result that would be computed by the numerical scheme.

(5 marks)

Question 4

Figure Q4 shows linearly varying distributed lift acting on the wing of an aircraft.

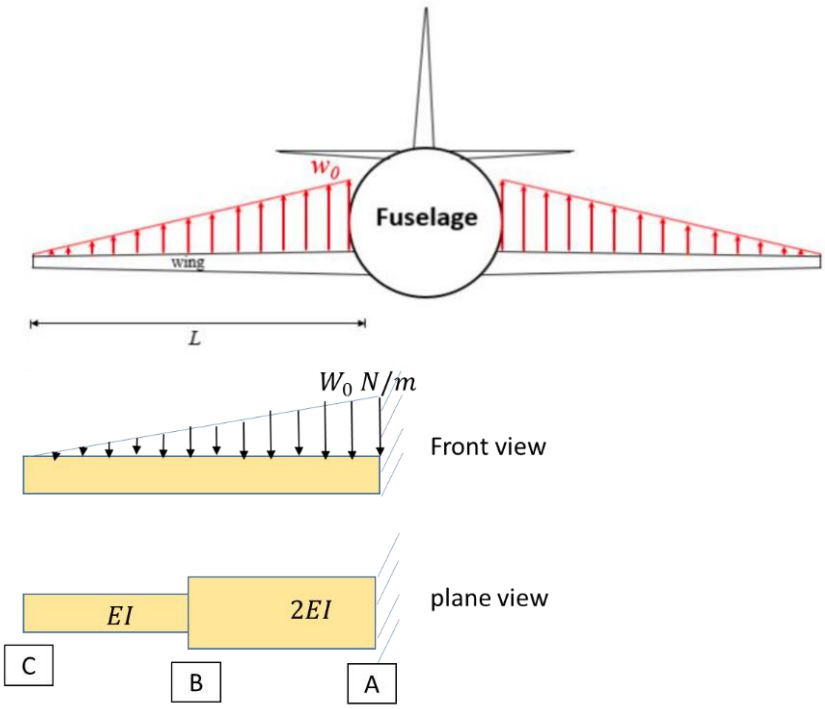

Figure Q4

A half span of the wing is represented by a beam with a fixed end supporting at the root.

Using Finite Element Method, the half span of the wing is represented by 2 beam elements.

The element stiffness matrix for a beam element is given by:

$$\begin{Bmatrix} f_{1y} \\ m_1 \\ f_{2y} \\ m_2 \end{Bmatrix} = \frac{EI}{L^3} \begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix} \begin{Bmatrix} v_1 \\ \phi_1 \\ v_2 \\ \phi_2 \end{Bmatrix}$$

Question 4a

The half wing is modelled using 2 beam elements. For the distributed lift, appraise the equivalent loads acting at nodes A, B and C, respectively.

(8 marks)

Question 4b

Compose the stiffness matrix of the entire half span wing.

(8 marks)

Question 4c

Design a suitable numerical procedure for the solution for the Reactions (bending moment and shear force) at node A. (Note: just provide the procedure, NO actual working is required.)

(9 marks)

----- END OF PAPER -----